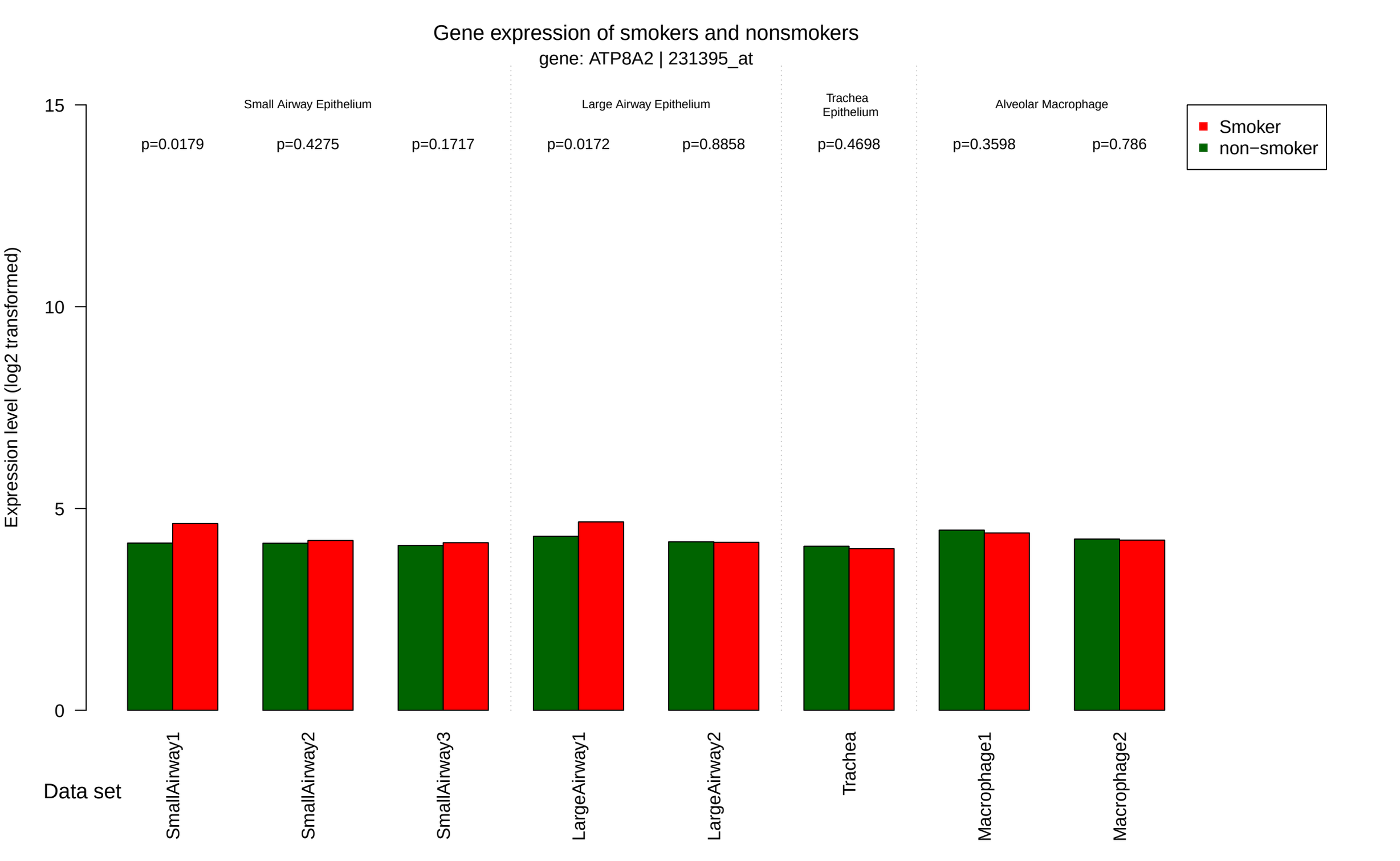
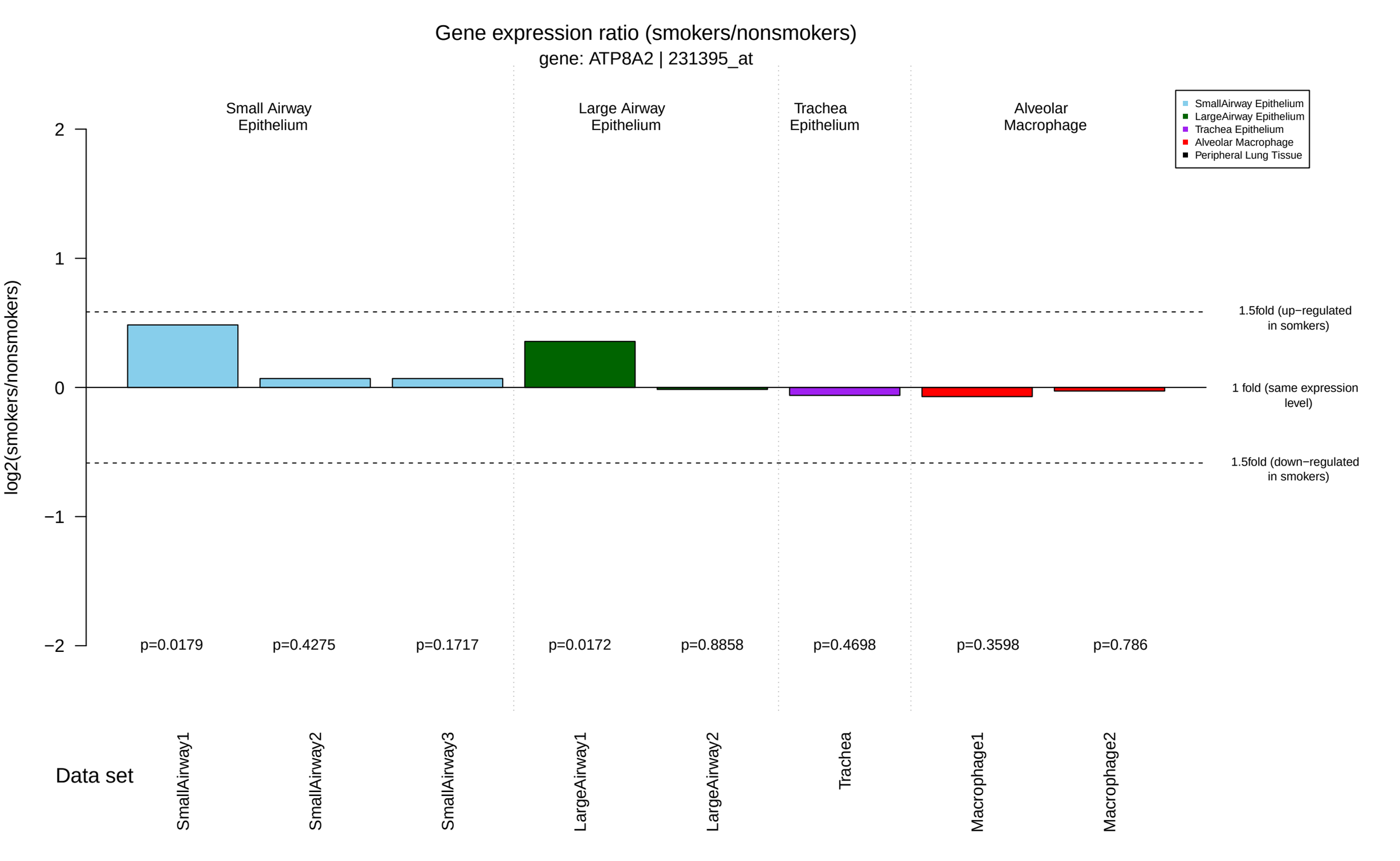
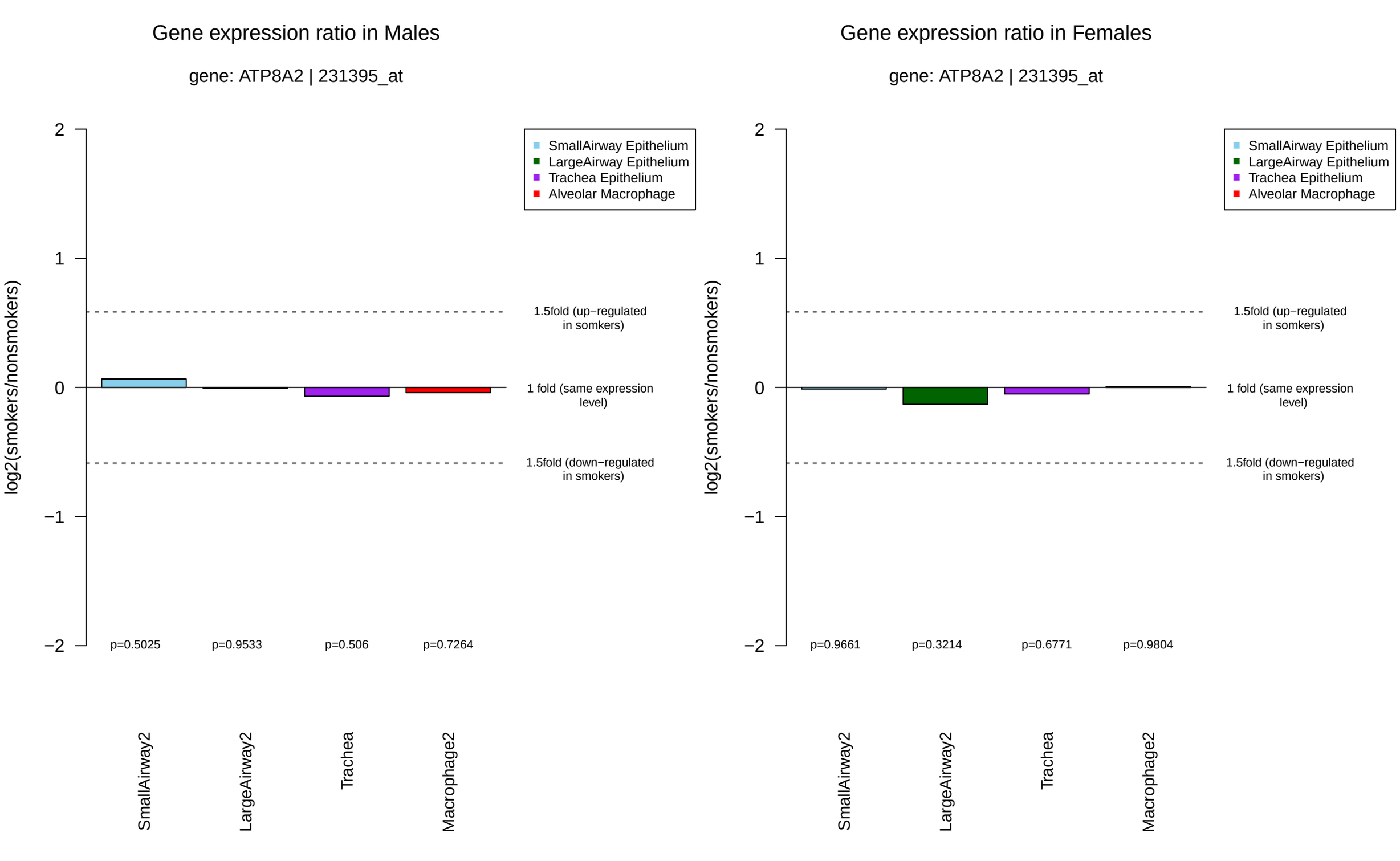
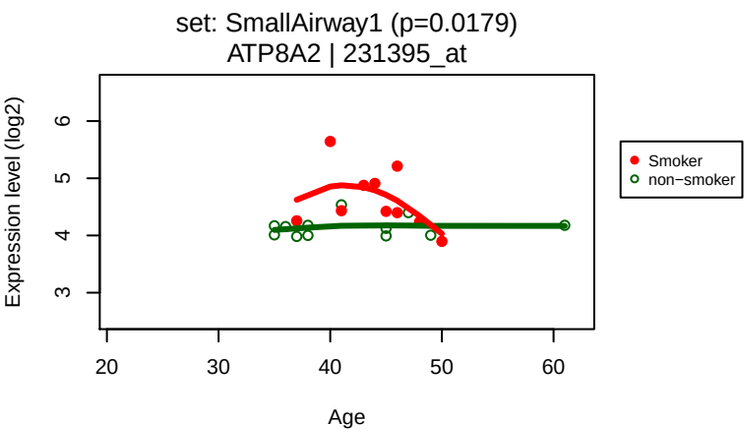
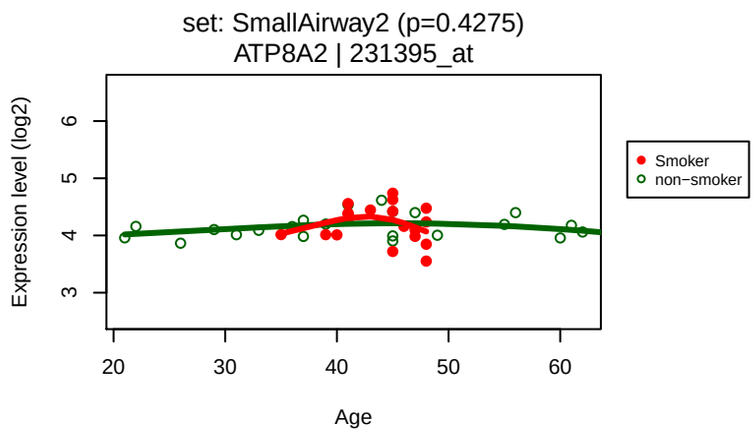
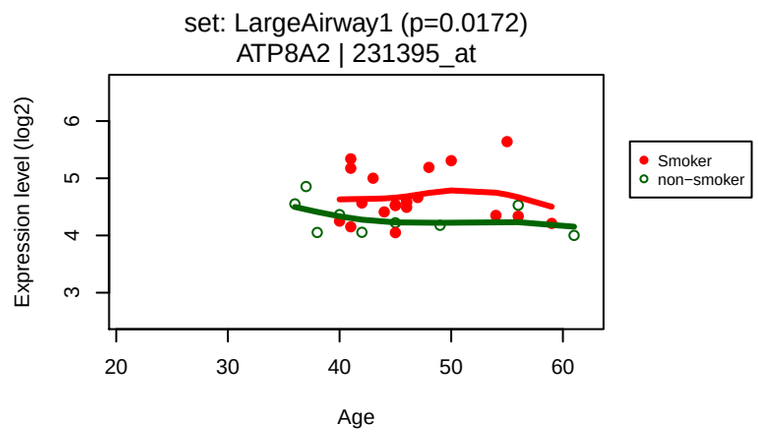

Gene expression of smokers and nonsmokers
gene: ATP8A2 | 231395_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0179
p=0.4275
p=0.1717
p=0.0172
p=0.8858
p=0.4698
p=0.3598
p=0.786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

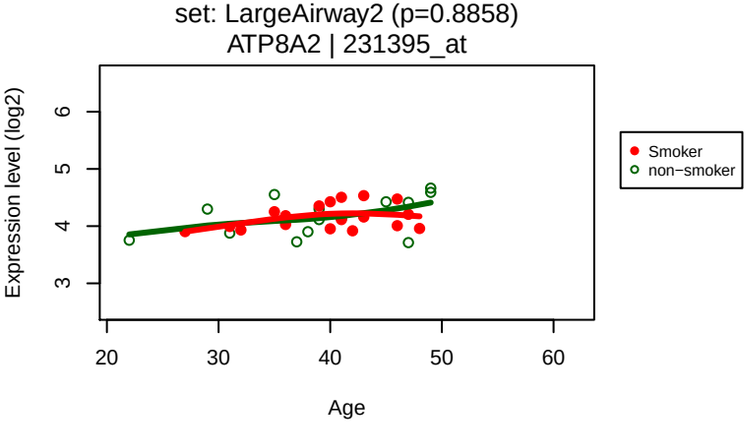

Gene expression ratio (smokers/nonsmokers)
gene: ATP8A2 | 231395_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0179
p=0.4275
p=0.1717
p=0.0172
p=0.8858
p=0.4698
p=0.3598
p=0.786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

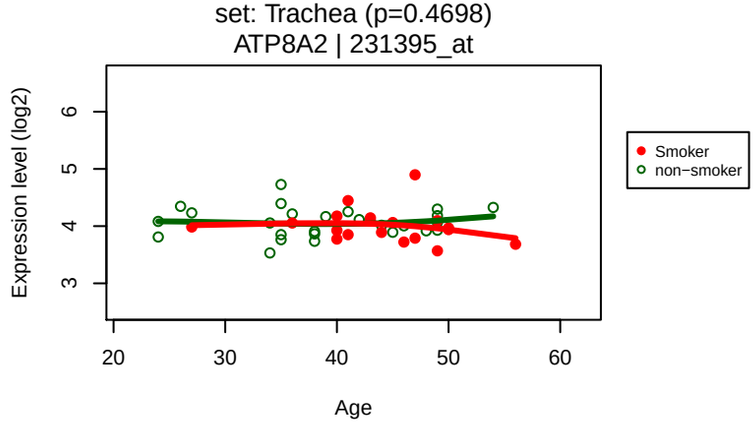

Gene expression ratio in Males
gene: ATP8A2 | 231395_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5025
p=0.9533
p=0.506
p=0.7264
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP8A2 | 231395_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9661
p=0.3214
p=0.6771
p=0.9804
SmallAirway2
LargeAirway2
Trachea
Macrophage2

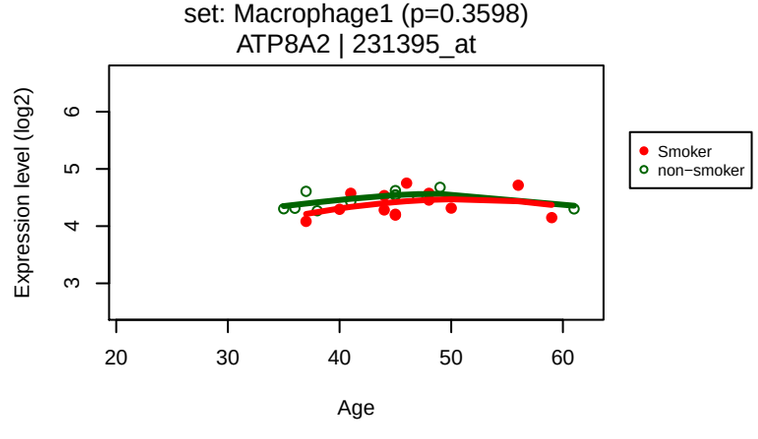

set: SmallAirway1 (p=0.0179)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker

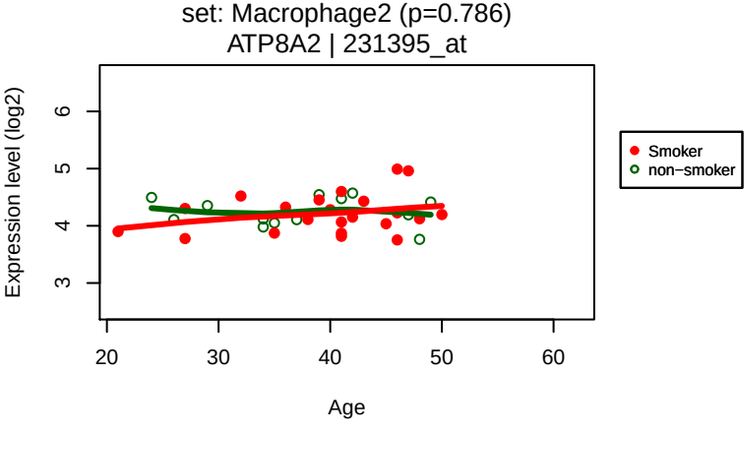

set: SmallAirway2 (p=0.4275)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker

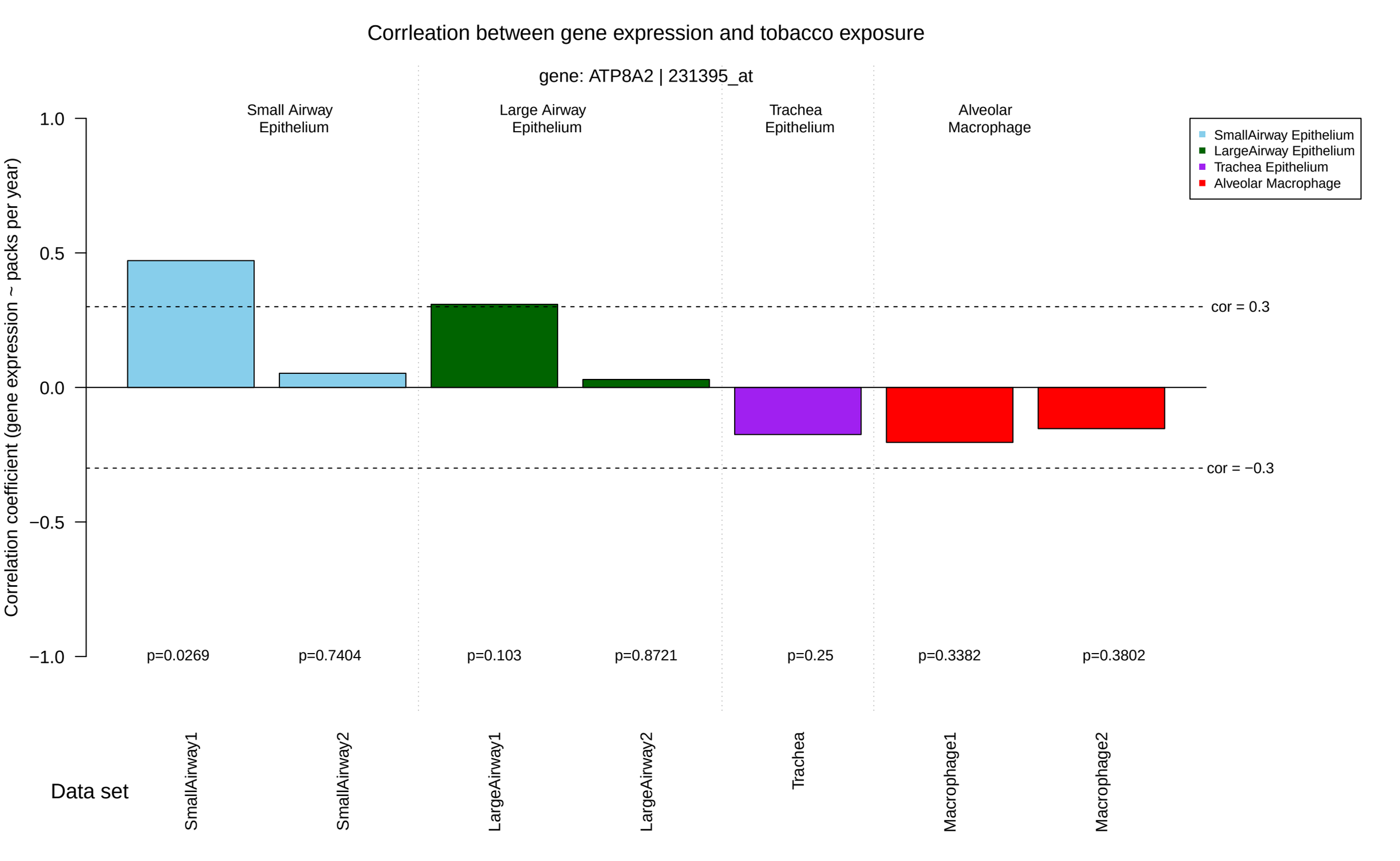

set: LargeAirway1 (p=0.0172)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8858)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4698)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3598)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.786)
ATP8A2 | 231395_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP8A2 | 231395_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0269
p=0.7404
p=0.103
p=0.8721
p=0.25
p=0.3382
p=0.3802
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2